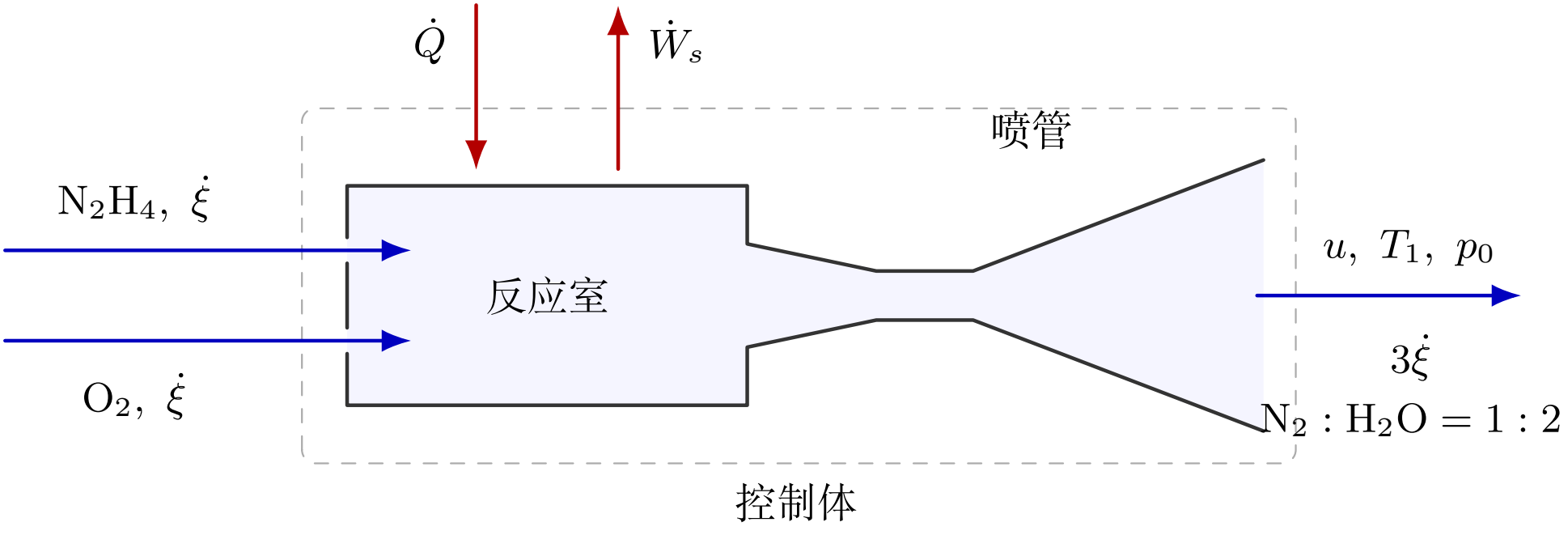
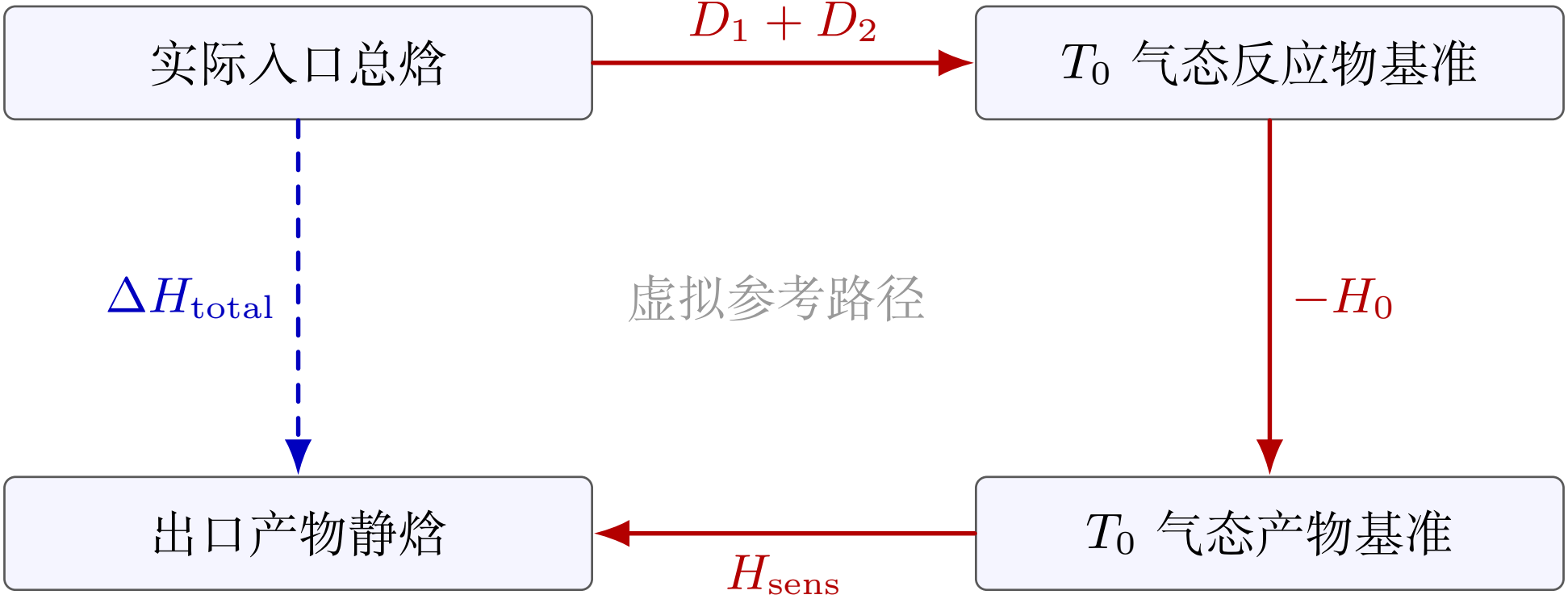
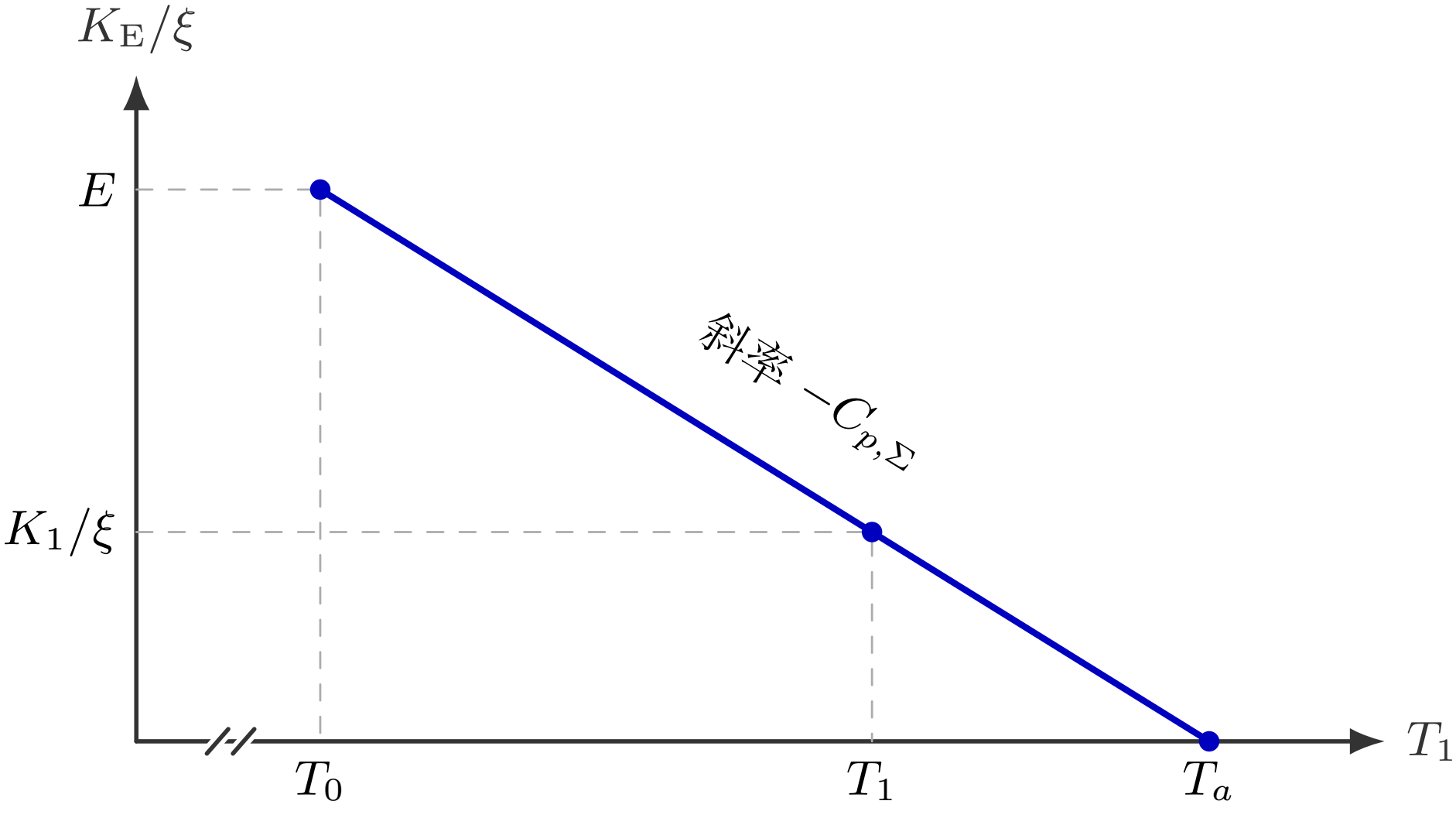
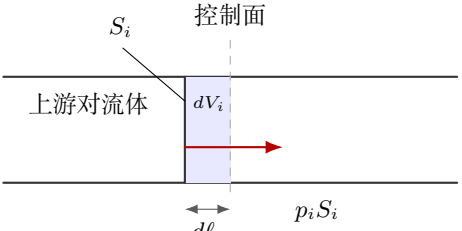
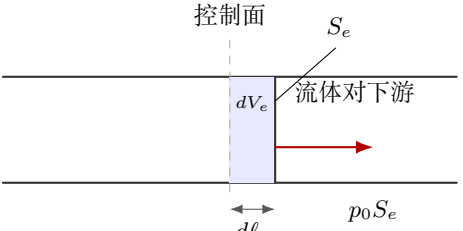

$\dot{Q}$
$\dot{W}_s$
喷管
$N_2H_4,\ \dot{\xi}$
反应室
$u,\ T_1,\ p_0$
$O_2,\ \dot{\xi}$
$3\dot{\xi}$
$N_2 : H_2O = 1 : 2$
控制体


实际入口总焓
$D_1 + D_2$
T_0 气态反应物基准
ΔH_{total}
虚拟参考路径
$-H_0$
出口产物静焓
H_{sens}
T_0 气态产物基准

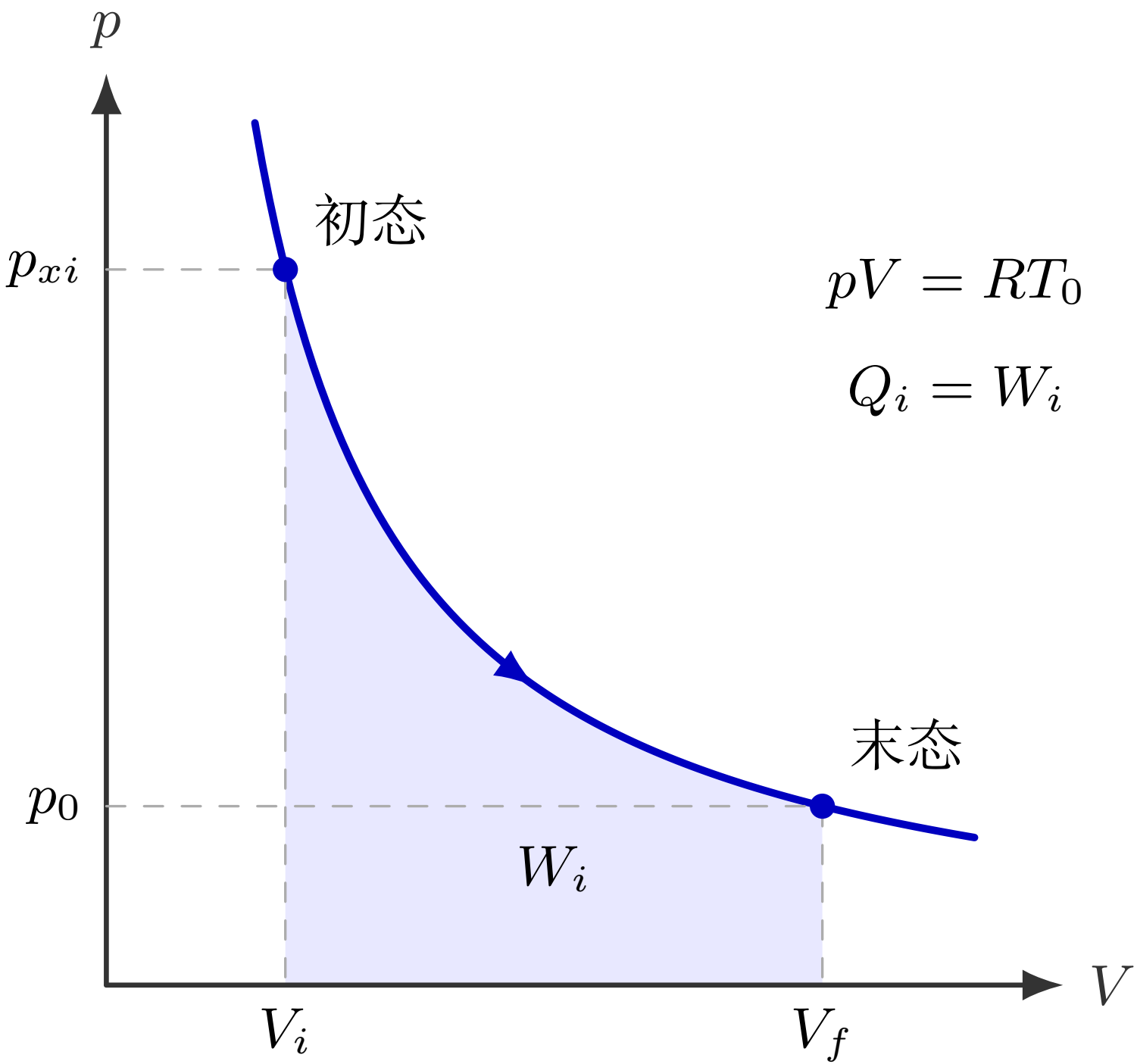

K_{E}/ξ
E
K_1/ξ
斜率 $-C_{p,\Sigma}$
T_1
T_0
T_1
T_a

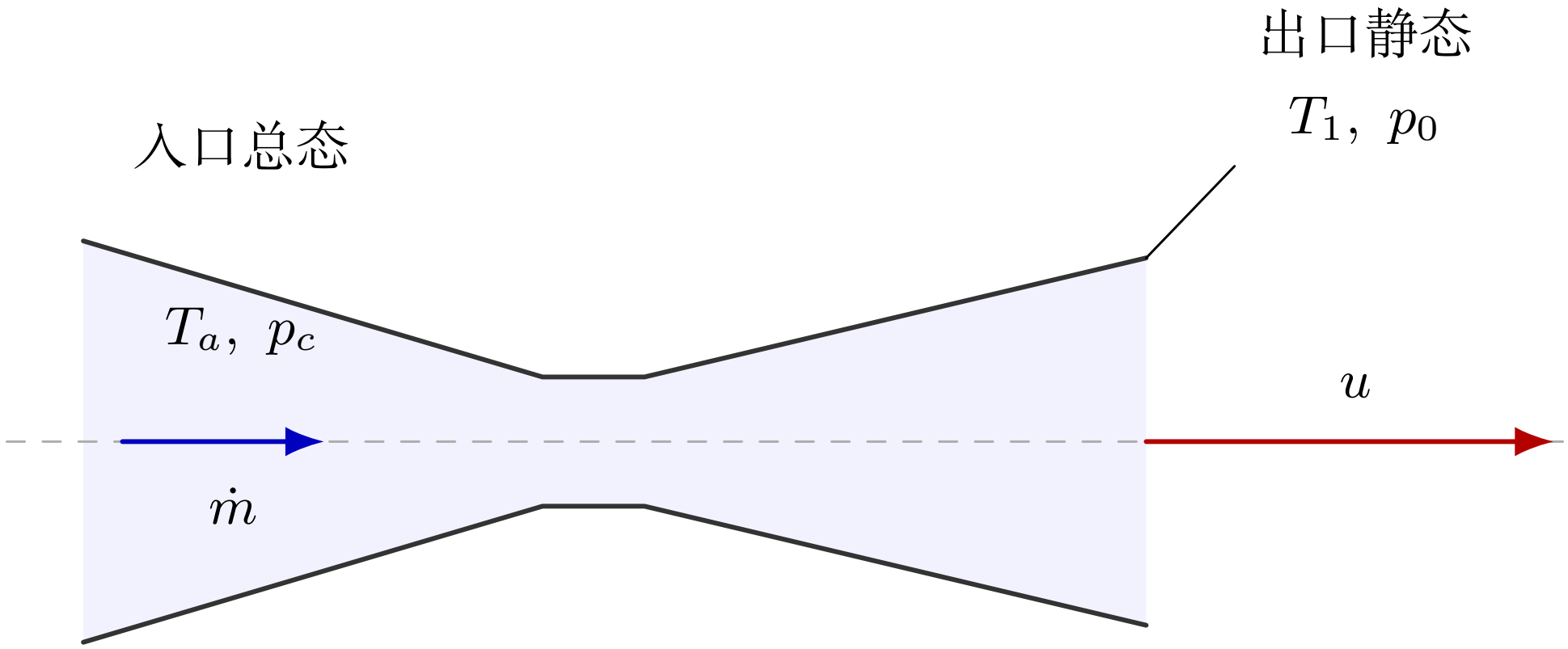


入口：压力功进入



出口：压力功流出


p
初态
p_{xi}
$pV = RT_0$
$Q_i = W_i$
末态
p_0
W_i
V
V_i
V_f



冻结组分的喷管模型